

# Nuclear magnetic resonance (NMR) spectroscopy

# What is NMR?

- **NMR (Nuclear Magnetic Resonance) spectroscopy is a powerful analytical technique that utilizes the magnetic properties of atomic nuclei to provide detailed information about the structure and composition of molecules.**
- **It works by exposing a sample to a strong magnetic field and radio waves, causing certain nuclei to absorb specific frequencies of radiation, which can then be used to determine the structure and properties of the molecule.**

# Principles

- Radiation in the radiofrequency region is used to excite atoms, like **protons or carbon-13 atoms**, so that their spins switch from being aligned with to being aligned against an applied magnetic field.
- The **range of frequencies** required for excitation and the **complex splitting patterns** produced are very characteristic of the **chemical structure of the molecule**.

- B/c of its charge & spin, a nucleus can behave like a magnet.
- NMR operates by applying  $B^0$  & then measuring the amount of energy necessary to put various nuclei in resonance.
- Nuclei in d/t electronic env'ts (shielded/deshielded) require d/t amounts of energy to bring them into resonance.
- An NMR spectrum provides a signal/peak representing the energy necessary to bring each nuclei into resonance.

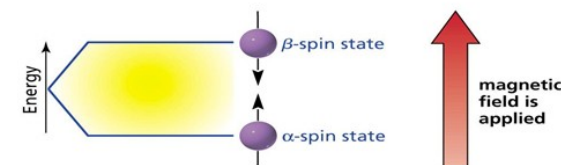
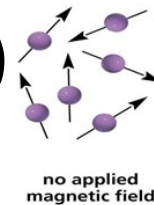
## Principle...

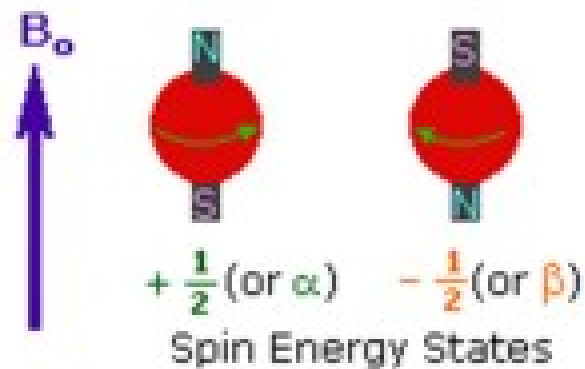
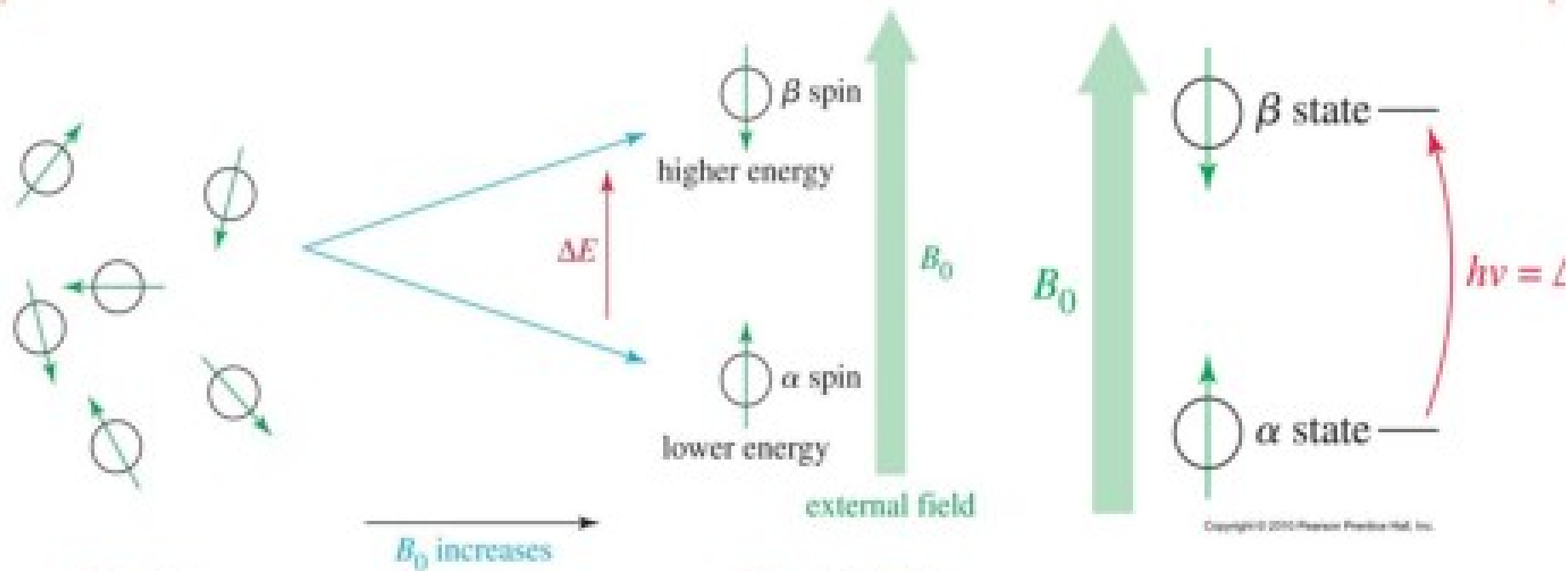
- A nucleus with spin charged nucleus behaves like a **tiny bar magnet**(generates a magnetic field).
- All nuclei with unpaired protons or neutrons are magnetically active.
- They have a magnetic field arising from the unpaired nuclear particle.
- However, inactive nuclei have an even # No of protons & neutrons that pair up with antiparallel spins giving  $I = 0$  (no net spin) i.e no angular momentum/magnetic mov't.

**NB: unpaired electrons, protons & neutrons each has a spin of  $1/2$**

## Principle...

- In the absence of an external magnetic field, the individual nuclei in a compound *are randomly oriented*.
- When an external magnetic field is applied to the nuclei:
  - Nuclei with spin  $+1/2$  orient themselves so that their tiny magnetic fields are in the same direction as the applied magnetic field (with the field)
  - Nuclei with spin  $-1/2$  orient opposite to the direction of the applied magnetic field (against the field)





A photon with the right amount of energy can be absorbed & cause the spinning proton to flip.

## Principle...

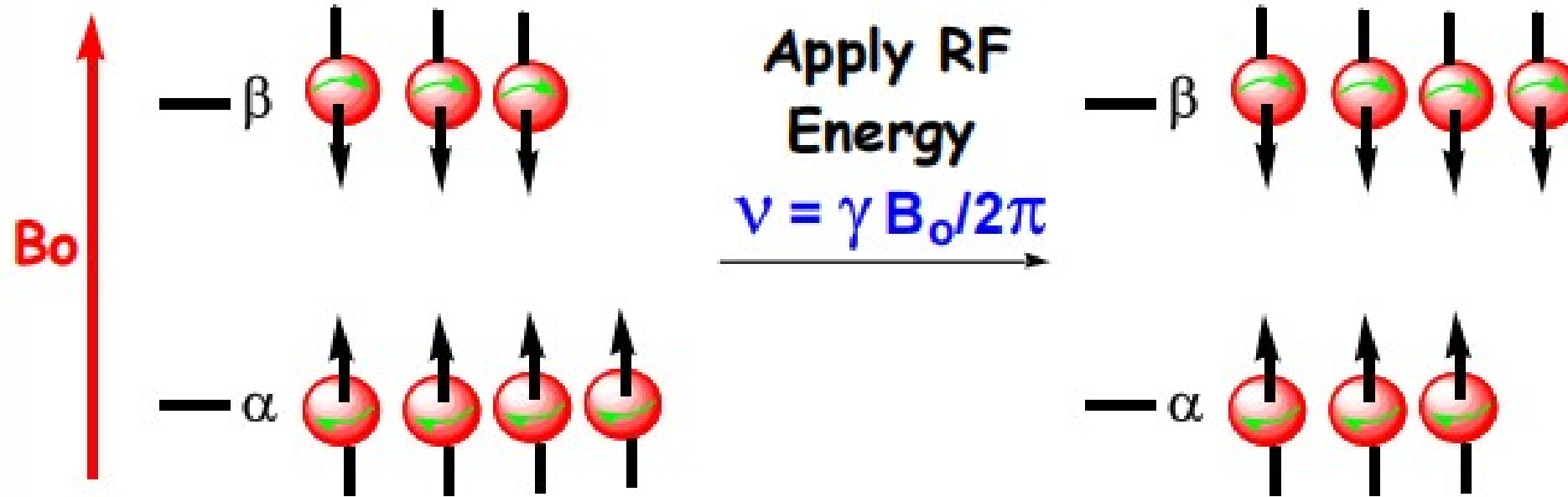
- Nuclei **aligned** with the field (*the  $\alpha$ -spin state*) have **lower energy** than those **aligned against** the field (*the  $\beta$ -spin state*)
  - The difference in the number of nuclei in the  $\alpha$ -spin and  $\beta$ -spin states is great
- For every million protons (hydrogen nuclei) there are some 10-20 more in the lower state than in the higher-energy state at room temperature
  - *The difference forms the basis of NMR spectroscopy*

## Principle...

- If EMR of the **appropriate energy** is applied to nuclei that have been oriented by a magnetic field:
  - *A nucleus in the  $\alpha$ -spin state will **absorb** the radiation **flipping its spin** and **entering the  $\beta$ -spin state***
    - **The absorption is detected, resulting in a signal in the NMR spectrum**
  - Energy absorbed in this transition equals **the energy difference ( $\Delta E$ )** between the  $\alpha$ - and  $\beta$ -spin states
  - The radiation required is in the radiofrequency (rf) region of the electromagnetic spectrum and is called **rf radiation**
  - When the nuclei undergo relaxation i.e., return to their original state ( $\alpha$ -spin state), they emit electromagnetic signals (**releasing energy as heat**) whose frequency depends on  $\Delta E$
  - “Nuclear Magnetic Resonance” indicates that the nuclei are in resonance with the EMR

*That is, they flip back & forth b/n the  $\alpha$ - and  $\beta$ -spin states in response to the EMR*

- APPLICATION OF RF ENERGY - FLIPPING



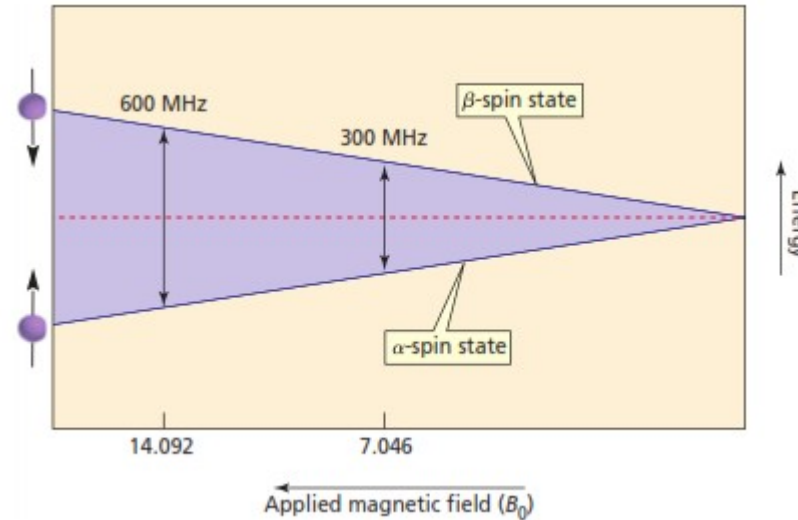
- In this context, “**resonance**” refers to the **flipping back & forth** of nuclei between **the  $\alpha$ - &  $\beta$ -spin states** in response to the RF radiation.

$\gamma$  - gyromagnetic ratio

## Principle...

- Energy difference between the  $\alpha$ - and  $\beta$ -spin states depends on the strength of the externally applied magnetic field
- The greater the  $H_0$ , the greater  $\Delta E$

$$\Delta E = h\nu = h \frac{\gamma H_0}{2\pi}$$



$\Delta E = 0$  MHz required for  $H_0 = 0$  Tesla

$\Delta E = 60$  MHz for  $H_0 = 1.409$  T

$\Delta E = 100$  MHz for  $H_0 = 2.350$  T

$\Delta E = 300$  MHz for  $H_0 = 7.046$  T

$\Delta E = 600$  MHz for  $H_0 = 14.09$  T

- The NMR spectrometer detects these signals and displays them as a plot of signal frequency versus intensity—an **NMR spectrum**
- It is because the nuclei are **in resonance** with the rf radiation that the term “nuclear magnetic resonance” came into being
- “Resonance” refers to the flipping back and forth of nuclei between the  $\alpha$ - and  $\beta$ -spin states in response to the rf radiation

## Principle...

The **gyromagnetic ratio** (also called the **magnetogyric ratio**)

- The ratio of the magnetic moment of a rotating charged particle to its angular momentum
- $\gamma$  depends is a constant that depends on the magnetic moment of the particular kind of nucleus
- For a proton  $\gamma$  is  **$2.675 \times 10^8$  radians  $\text{T}^{-1}\text{s}^{-1}$** ; in the case of a  $^{13}\text{C}$  nucleus, it is  **$6.688 \times 10^7$   $\text{T}^{-1}\text{s}^{-1}$**

Example: If NMR spectrometer is equipped with a magnetic field of 1.4092 T, the spectrometer will require an energy source of a  $\nu$  of:

$$\Delta E = h\nu = h \frac{\gamma H_0}{2\pi} \implies \nu = \frac{\gamma H_0}{2\pi} = \frac{2.675 \times 10^8 \text{ radians } \text{T}^{-1}\text{s}^{-1} \times 1.4092 \text{ T}}{2 \times 3.1416} = 60 \times 10^6 \text{ s}^{-1} = 60 \text{ MHz}$$

# Question

- What frequency (in MHz) is required to cause a proton to flip its spin when it is exposed to a magnetic field of 1 T?
- Calculate the magnetic field (in tesla) required to flip an  $^1\text{H}$  nucleus in an NMR spectrometer that operates at 360 MHz.
- What strength magnetic field is required when a 500-MHz instrument is used for  $^1\text{H}$  NMR?

## Principle...

- More **powerful magnet** requires energy source with a **higher frequency**
  - E.g. when  $H_0$  is 2.3486, an energy source with a frequency of 100 MHz is required
- **Operating frequency depends on the strength of the built-in magnet**
  - The greater the operating frequency the ***stronger the magnet*** & the ***better the resolution*** of the NMR spectrum
  - For different nuclei, different energy sources are required to bring the nuclei **into resonance** in a spectrometer with a given operating frequency
    - E.g. an NMR spectrometer having a magnet requiring an energy source of 60 MHz to flip the spin of  $^1\text{H}$  nucleus requires an energy source of 15 MHz to flip the spin of a  $^{13}\text{C}$  nucleus
    - Many NMR spectrometers are equipped with radiation sources that can be **tuned to different frequencies**

## Nuclei that can be studied by NMR

- In a nuclide represented by  ${}^A_Z\text{X}$ , the nucleus will have a non-zero spin number  $I$ , if the number of protons  $Z$  and the number of neutrons  $A$ , are **not both even numbers**.
- ${}^1_1\text{H}$ ,  ${}^{13}_6\text{C}$ ,  ${}^{19}_9\text{F}$ ,  ${}^{31}_{15}\text{P}$  have, for example, a spin number  $I = 1/2$  while  $I = 1$  for  ${}^2_1\text{H}$  (deuterium D) or  ${}^{14}_7\text{N}$ .
  - All these nuclei will give an NMR signal.
- Alternatively, the nuclei  ${}^{12}_6\text{C}$ ,  ${}^4_2\text{He}$ ,  ${}^{16}_8\text{O}$ ,  ${}^{28}_{14}\text{Si}$ ,  ${}^{32}_{16}\text{S}$  which have **a spin number of zero, cannot be studied by NMR**.
  - **NB. Odd mass, odd atomic number, or both has quantized spin angular momentum & a magnetic moment** (e.g.  ${}^1\text{H}$ ,  ${}^{13}\text{C}$ ,  ${}^{14}\text{N}$ ,  ${}^{17}\text{O}$ ,  ${}^{19}\text{F}$ )

# NMR INSTRUMENT

## **4 components:**

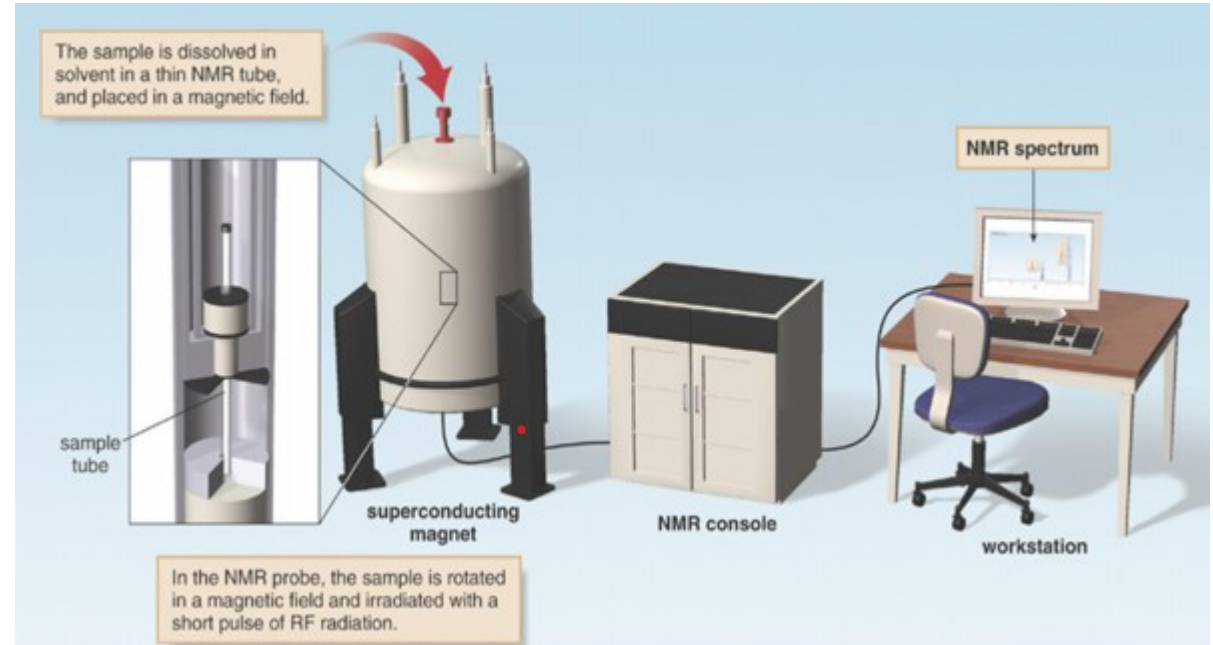
**1. NMR powerful magnet**

**2. NMR probe**

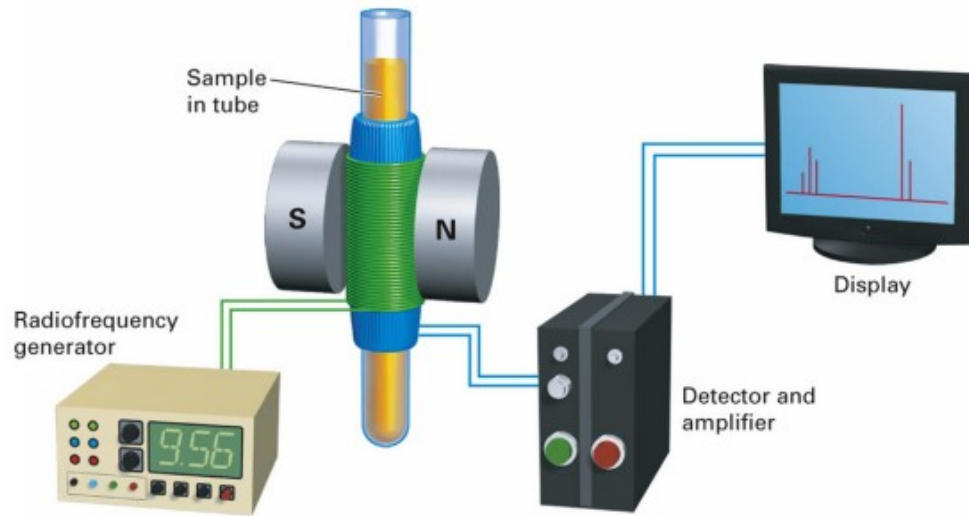
**3. NMR console**

- **Radio-frequency generator**
- **Radio-frequency detector**
- **Signal amplifier**

**4. Computer workstation & control software (data processing)**



# NMR INSTRUMENT



- A radio transmitter coil that produces a short powerful pulse of radio waves.
- A powerful magnet that produces strong magnetic fields
- The sample is placed in a glass tube that spins so the test material is subject to uniform magnetic field.
- The sample is dissolved in deuterated solvents to eliminate  $^1\text{H}$  signals from the solvent.
- A radio receiver coil that detects radio frequencies absorbed by nuclei.

# How to take NMR spectrum?

- To obtain an NMR spectrum, one dissolves a small(5-50 mg) amount of a compound containing hydrogen atoms in about 0.5 mL of solvent and **puts the solution into a long, thin glass tube, which is placed within a powerful magnetic field “with a variable strength”** .
- **Spinning the sample** tube about its long axis averages the position of the molecules in the magnetic field.
- **The individual protons absorb the frequency required to come into resonance (flip their spin).**

# How to take NMR spectrum?

- As the protons relax (i.e., as they return to equilibrium), they produce a **tiny electrical current to flow** in an antenna coil surrounding the sample.
- The instrument then **amplify this current** and display it as a signal (**NMR Spectrum**) on a recorder.

## Shielding and Deshielding of Protons

- If all the protons in an organic compound were **in exactly the same environment**, they would all give signals with the same *frequency in response to a given applied magnetic field*.
- If this were the case, ***all NMR spectra would consist of one signal***, which would tell us ***nothing about the structure*** of the compound, except that it contains protons.
- A nucleus, however, is embedded in a cloud of electrons that partly ***shields*** it from the applied magnetic field.

## Shielding and deshielding...

- A nucleus (protons) is *embedded in a cloud of e<sup>-</sup>s that partly shields it from H<sub>0</sub>*
  - Shielding varies for different protons within a compound
  - **Spinning e<sup>-</sup>s induce local magnetic field (H<sub>local</sub>) that opposes (i.e. subtracts from) H<sub>0</sub>**
- Therefore,
  - The effective magnetic field (EMF; H<sub>effective</sub>) is what the nuclei “sense” through the surrounding magnetic environment:
    - **$H_{\text{effective}} = H_{\text{applied}} - H_{\text{local}}$**
  - The EMF is the one that the proton “senses”
  - Protons that are **more shielded** require **stronger H<sub>0</sub>** in order to “sense” a given EMF
  - **Less shielded** protons require **weaker H<sub>0</sub>** in order to “sense” the same EMF
- The above type of shielding is known as **diamagnetic shielding**

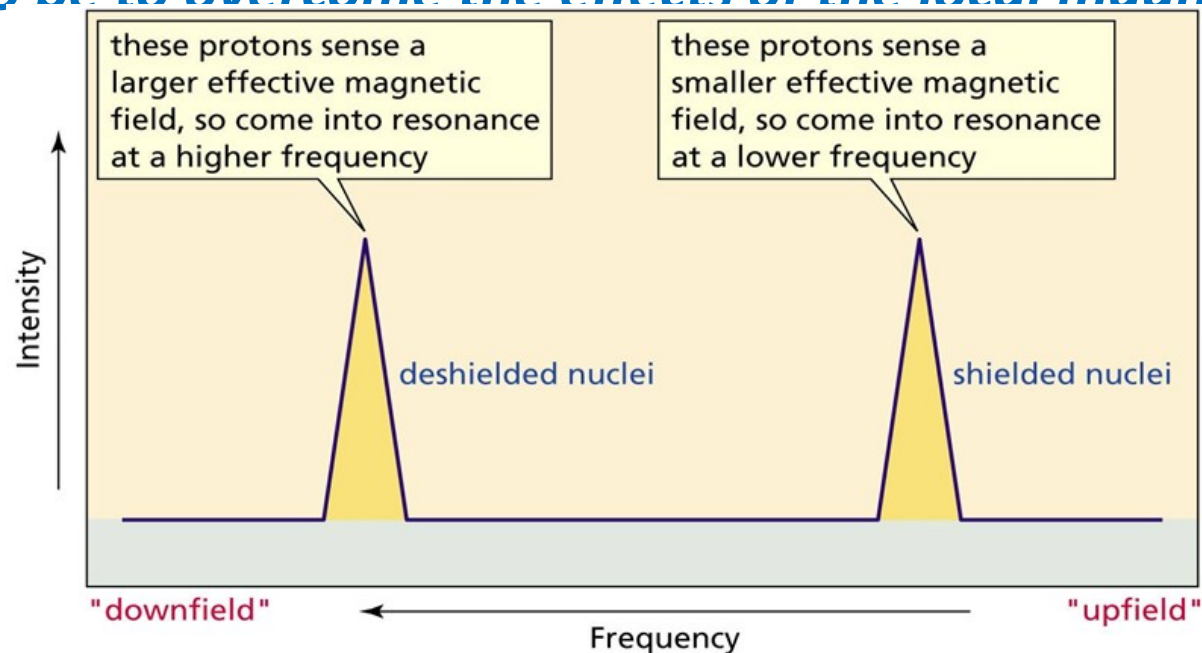
## Shielding and deshielding...

- **Thus,**
  - **Hs in e<sup>-</sup>-dense environments sense a smaller EMF**
    - Require a smaller frequency to come into resonance -i.e. flip their spins- because  $\Delta E$  is smaller
  - **Hs in e<sup>-</sup>-poor environments sense a larger EMF**
    - Require a higher frequency to come into resonance

## Shielding and deshielding...

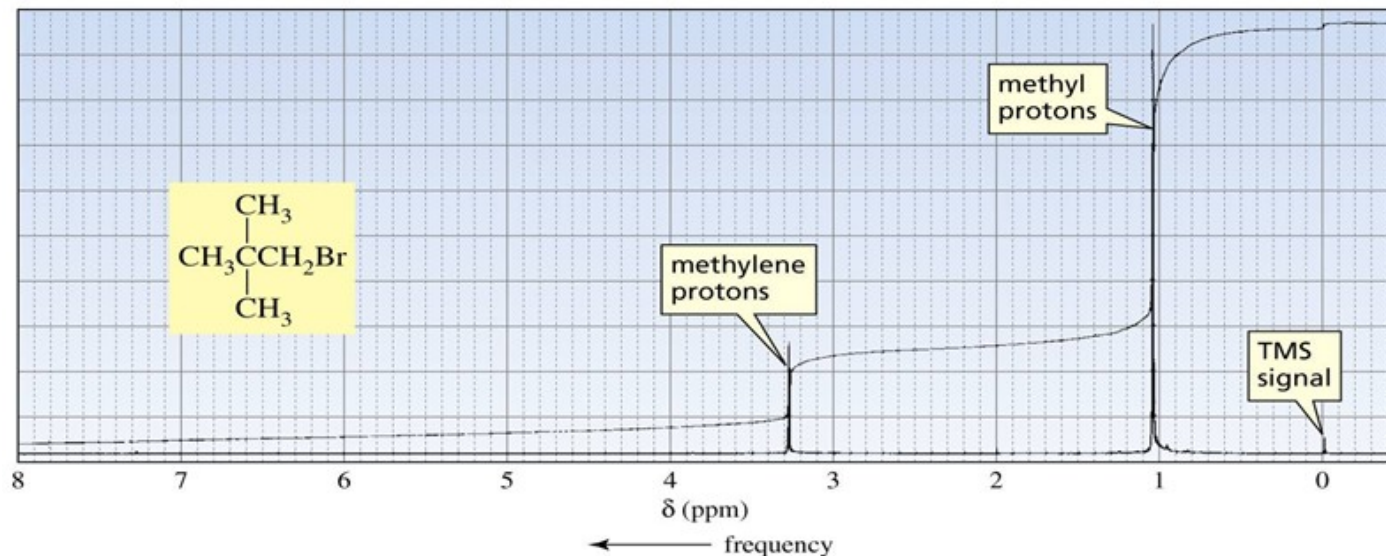
### Conclusion

- The greater the  $e^-$  density of the environment in which the proton is located the more the proton will be shielded from  $H_0$ , the greater the applied magnetic field will have to be to overcome the effects of the local magnetic field.*



## Shielding and deshielding...

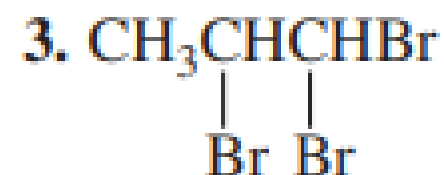
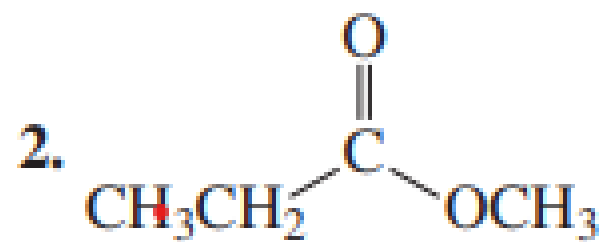
- Example:  $^1\text{H}$  NMR spectrum of 1-bromo-2,2-dimethylpropane



- Facts:
  - Because of shielding and deshielding 1-bromo-2,2-dimethyl propane has **2 different kinds of Hs**
    - $\text{CH}_2$  Hs are in a **less e- dense environment** than the  $\text{CH}_3$  Hs
    - $\text{CH}_2$  Hs are **closer** to the e- withdrawing bromine
      - Thus,  $\text{CH}_2$  Hs are less shielded from the  $\text{H}_0$

# PROBLEM

- a. Which set of protons in each of the following compounds is the least shielded?
- b. Which set of protons in each compound is the most shielded?

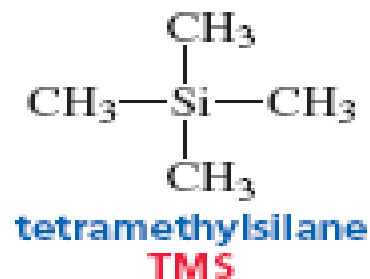


# **Chemical shift**

- Shielding and deshielding process cause the **absorption of protons to be shifted from the expected positions**
- Since these shifts result from the effect of electrons in chemical bonds, thus they are called **“Chemical Shifts”**
- In order to standardize the NMR scale it is necessary to set ***a 0 reference point*** to which all protons can then be compared.

## Chemical shift...

- The standard reference that was chosen is **tetramethylsilane (TMS)**.



- A small amount of TMS is usually added to sample whose spectrum is being measured, and the signal from its protons (**12 identical protons**) is used to **establish the zero point on the data scale.**

## **Chemical shift...**

### **Why TMS?**

- 1- It has 12 equivalent protons, therefore, a very small amount of TMS gives a relatively large signal.**
- 2- All hydrogens are identical , thus give a single signal**
- 3- Silicon is less electronegative than carbon, thus the protons of TMS are highly shielded than the analogous carbon compounds, and its signal occurs in a region of spectrum where few other H atoms absorb.**
- 4- Its signal seldom interferes with the signals from other hydrogen atoms.**
- 5- TMS is relatively inert compound.**
- 6- TMS is a volatile compound, its b.p. is 27 °C , thus after the spectrum has`been determined, TMS can be removed easily by evaporation.**

# Chemical shift ( $\delta$ )

- **Measured in parts per million**
  - Refers to position of signal in NMR spectrum
  - Indicates how far the signal is from TMS peak
  - Called the  $\delta$  (delta) scale
  - Determined by measuring the distance from TMS peak (in hertz) and dividing by the operating frequency of the instrument (in MHz)

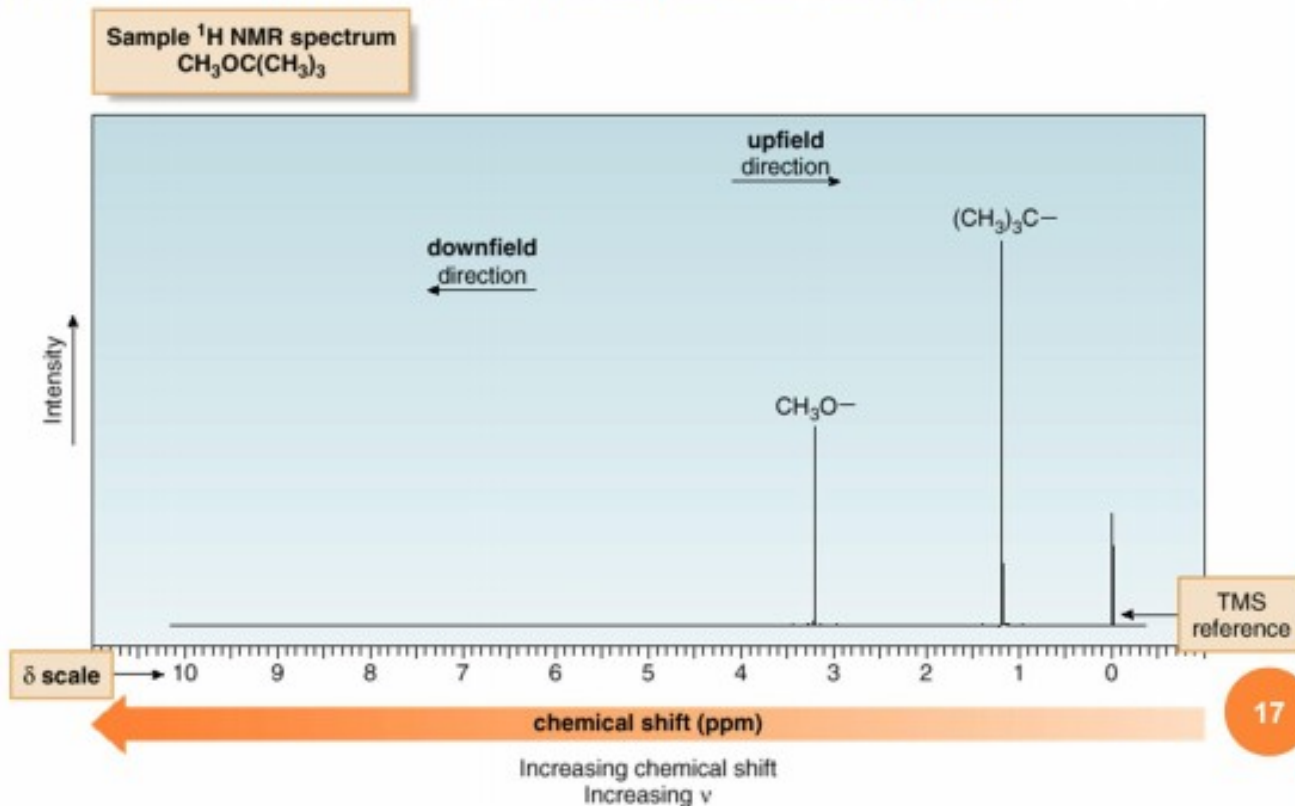
$$\text{Chemical shift } (\delta) = \frac{\text{Distance downfield from TMS (Hz)}}{\text{Operating frequency of the spectrometer (MHz)}}$$

**Thus:**

- In  $\delta$  scale TMS has a zero position
- High magnetic fields strengths (upfield) – low  $\delta$  values
- Low magnetic fields strengths (downfield) – large  $\delta$  values

# Chemical shift ( $\delta$ )

- An NMR spectrum is a plot of the intensity of a peak against its chemical shift, measured in parts per million (ppm).

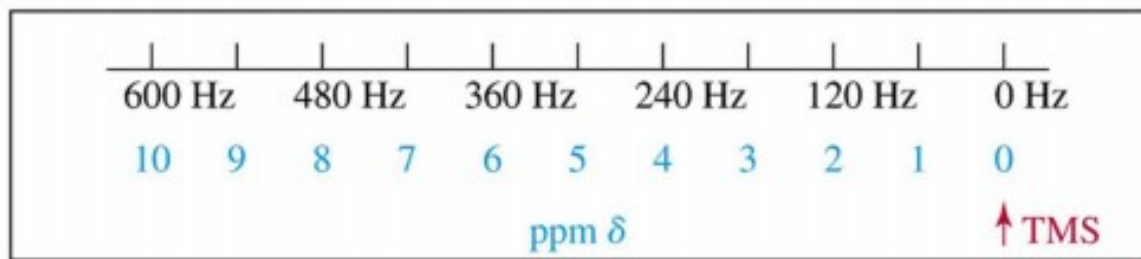


- NMR absorptions generally appear as sharp peaks.
- Most protons absorb b/n 0-10 ppm.
- The terms “upfield” & “downfield” describe the relative location of peaks.
  - Upfield means to the right.
  - Downfield means to the left

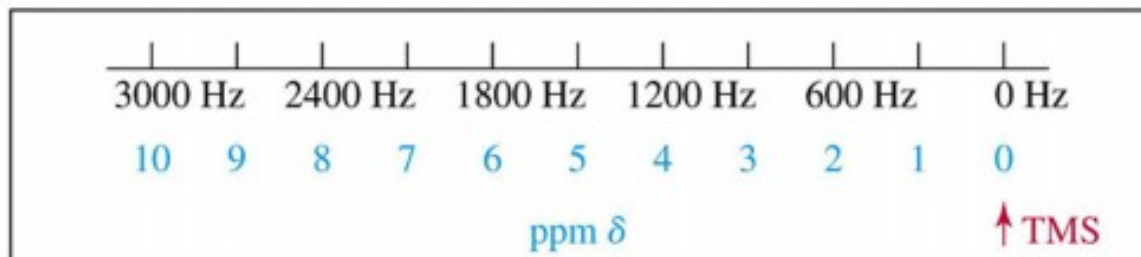
## Chemical shift ( $\delta$ )

- For a  $^1\text{H}$ -NMR spectrum, signals are reported by their shift from the 12 H signal in TMS.
- For a  $^{13}\text{C}$ -NMR spectrum, signals are reported by their shift from the 4 C signal in TMS.

□ *The advantage of the scale is that the chemical shift of a given nucleus is independent of the operating frequency of the NMR spectrometer.*



60 MHz



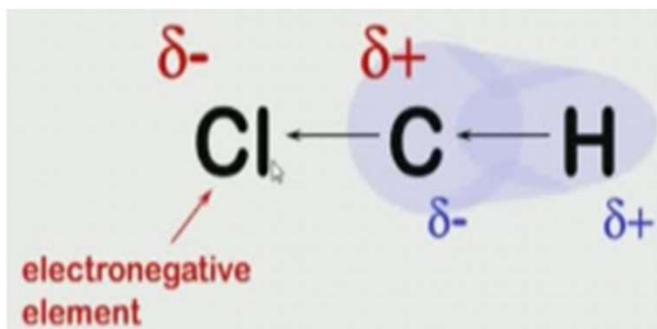
300 MHz

# **FACTORS INFLUENCING CHEMICAL SHIFT**

- ☐ **Inductive effect**
- ☐ **Hybridization of carbon**
- ☐ **Mesomeric /resonance effect**
- ☐ **Steric effect**
- ☐ **Hydrogen bonding**

# INDUCTIVE EFFECT

- Its effect on chemical shift is caused by  $e^-$  withdrawing or  $e^-$  donating groups attached to the H.
- Electron withdrawing groups such NH, OH,  $\text{COO}^-$ , Cl etc shift the H chemical shift to downfield regions.



- Chlorine deshields the proton.
  - It takes  $e^-$  density away from carbon, w/c in turn takes more density away from the proton.

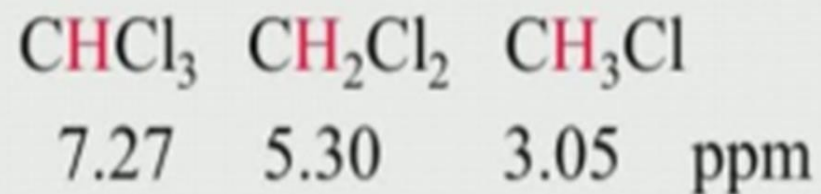
## INDUCTIVE EFFECT.....

- o Chemical shift increases with increasing electronegativity of X in methyl halides.

| CH <sub>3</sub> -X                 | Electronegativity of X | δ of H               |
|------------------------------------|------------------------|----------------------|
| CH <sub>3</sub> F                  | 4.0                    | 4.26                 |
| CH <sub>3</sub> OH                 | 3.5                    | 3.47                 |
| CH <sub>3</sub> Cl                 | 3.1                    | 3.05                 |
| CH <sub>3</sub> Br                 | 2.8                    | 2.68                 |
| CH <sub>3</sub> I                  | 2.5                    | 2.16                 |
| (CH <sub>3</sub> ) <sub>4</sub> C  | 2.1                    | 0.86                 |
| (CH <sub>3</sub> ) <sub>4</sub> Si | 1.8                    | 0.00 (by definition) |

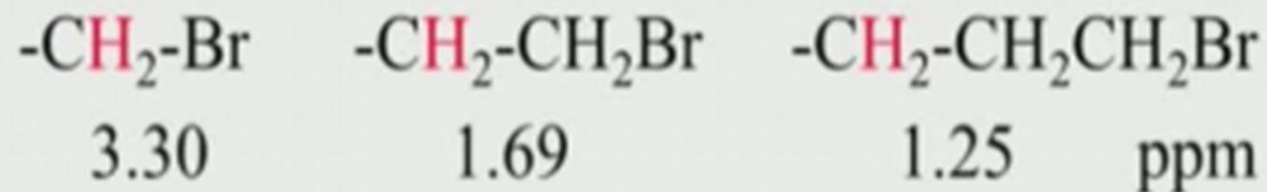
# INDUCTIVE EFFECT...

deshielding



The effect increases with greater numbers of electronegative atoms.

deshielding



The effect decreases with increasing distance.

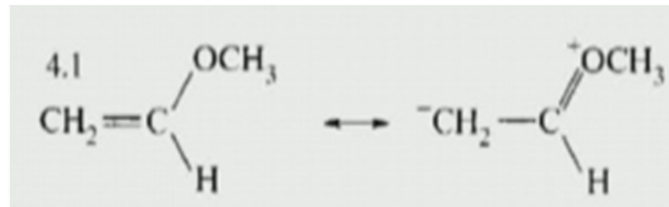
## HYBRIDIZATION OF CARBON ATOM ATTACHED TO H

- Hybridization of the carbon to w/c the proton is attached influences the e- density at H attached.
- As the proportion of 's' character ↑s from  $sp^3$  to  $sp$  orbitals,
  - Bonding electrons move closer to carbon & away from the protons, w/c become deshielded.
  - This is similar to the effect when electronegative atom is attached.

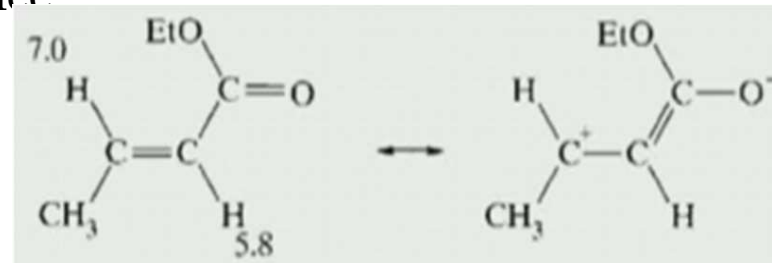
| $sp^3$ hybridization |          | $sp^2$ hybridization |          |
|----------------------|----------|----------------------|----------|
| $CH_4$               | 0.23 ppm | $CH_2=CH_2$          | 5.28 ppm |
| $CH_3-CH_3$          | 0.86 ppm |                      |          |

# MESOMERIC (RESONANCE) EFFECT

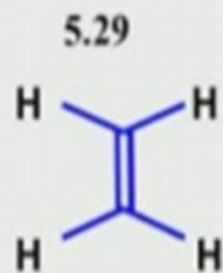
- As a result of resonance, the carbon atom accumulates a negative charge or becomes positively charged,
  - W/c influences the electron density around the hydrogen
    - A negative charge refers to accumulated higher electron density.



- The electron density around the hydrogen is reduced if carbon is positively charged.
  - H therefore appears at a very down field shifted value of a 7 ppm compared to if there was no resonance



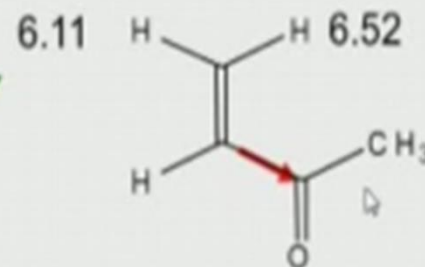
## Mesomeric effects



All the protons are equivalent  
and resonates at 5.29 ppm

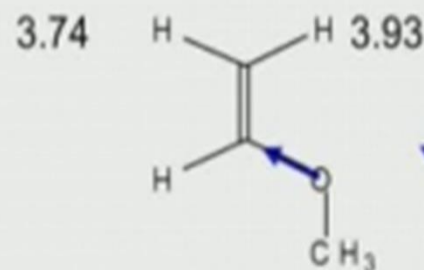
The addition of a ketone group withdraws electron  
and decreases the shielding around protons

### Methylvinylketone



Ethylene protons  $\delta$  shifts downfield

### Methylvinylether

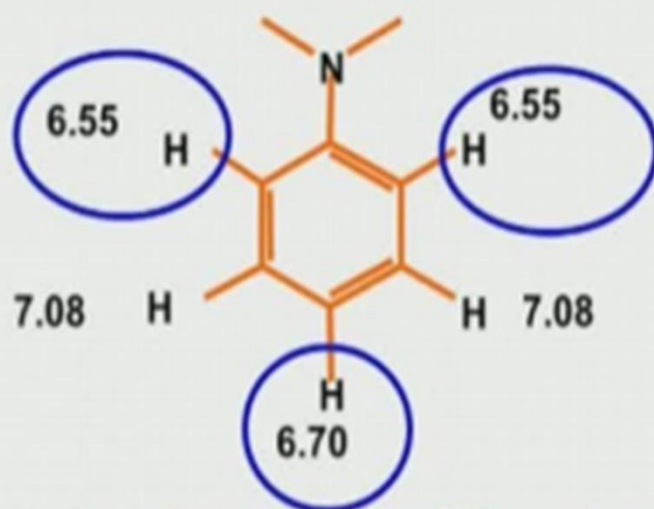


Methoxy group will donate electron to the double bond and increases  
the electron density around protons and increases shielding

resonates at high field

## Mesomeric effects

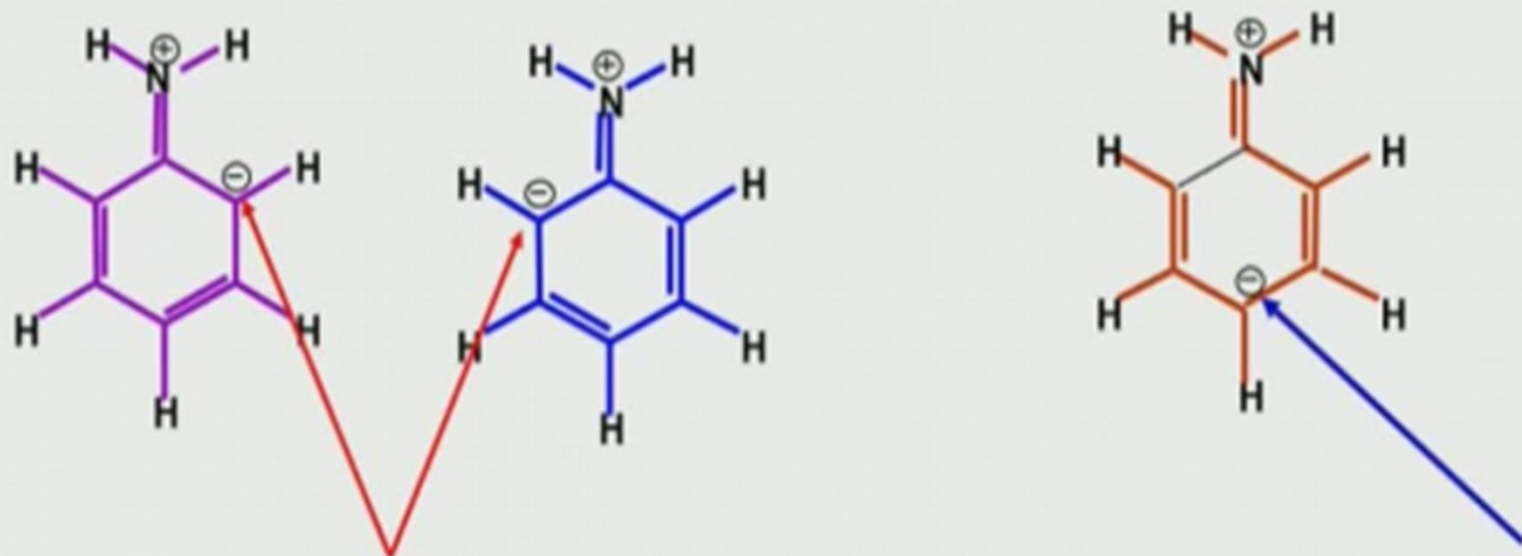
### Aniline



Electron donating group

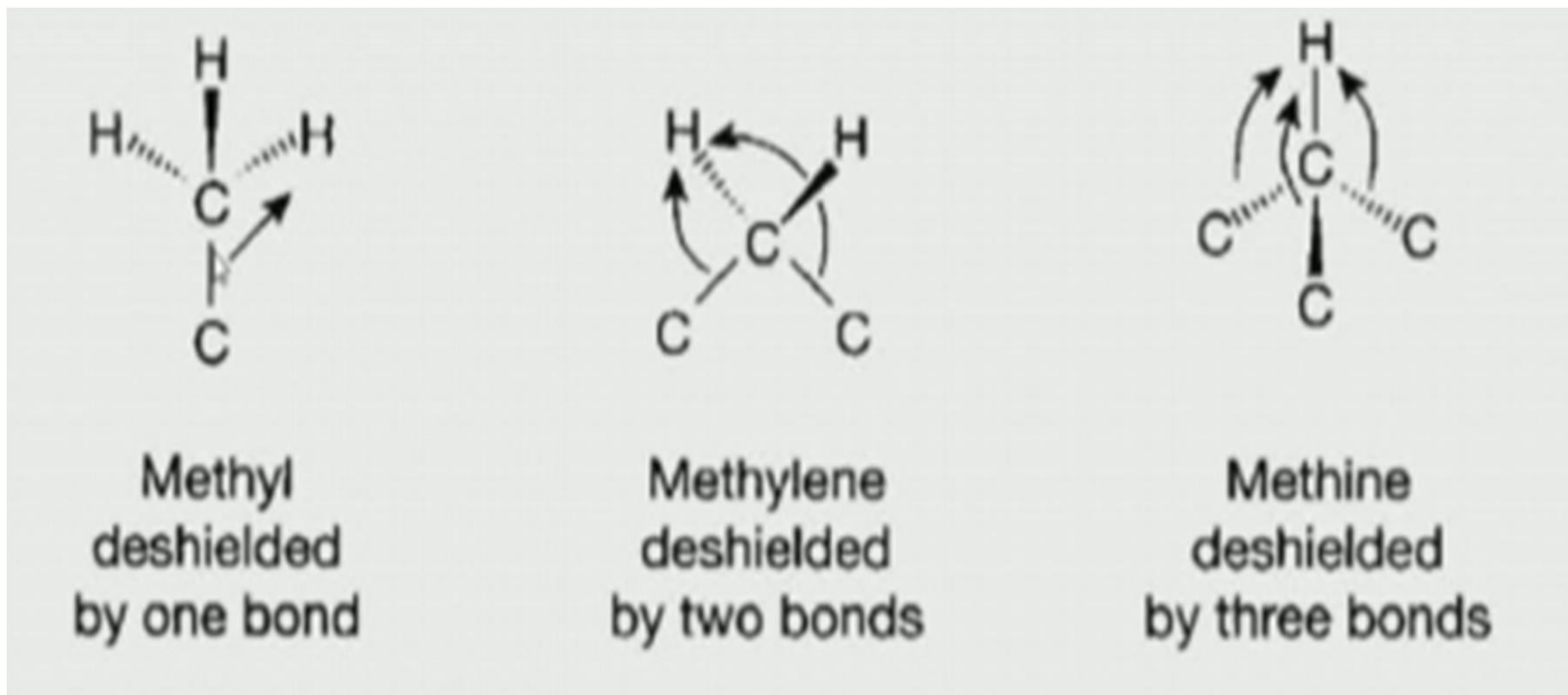
Much lower than benzene ring proton shifts (7.24 ppm)

Resonance structures of aniline shows that electron densities are more at ortho and para protons : results in lower chemical shifts



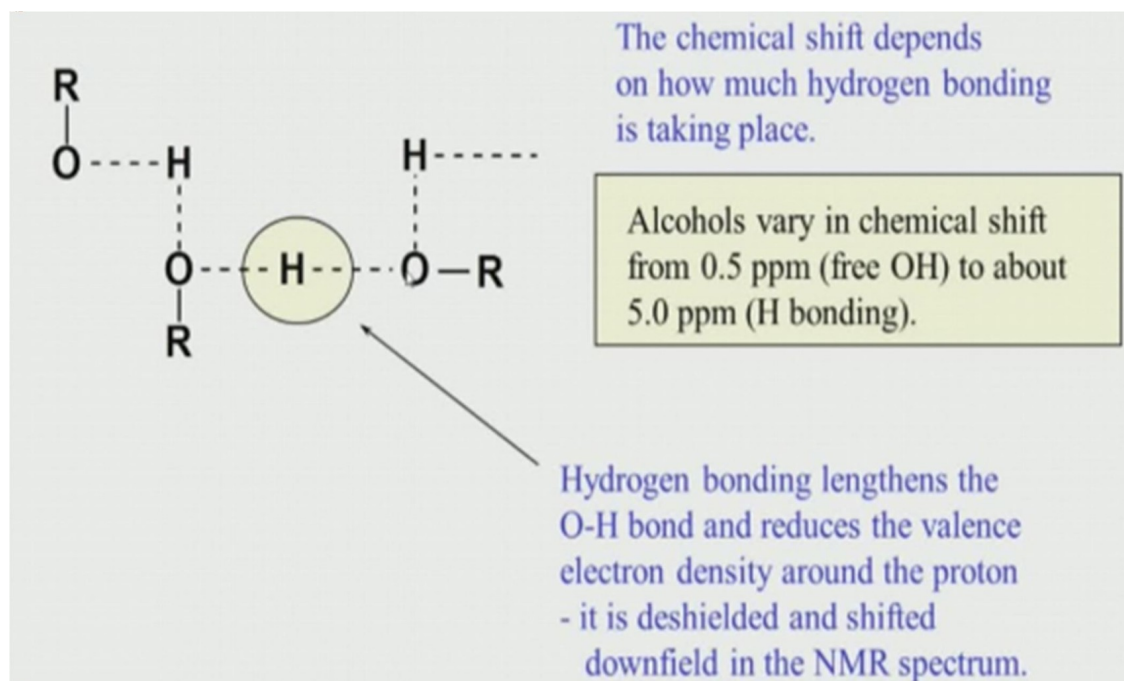
# STERIC EFFECT

- Steric crowding around a carbon nucleus increases deshielding & moves resonances downfield.

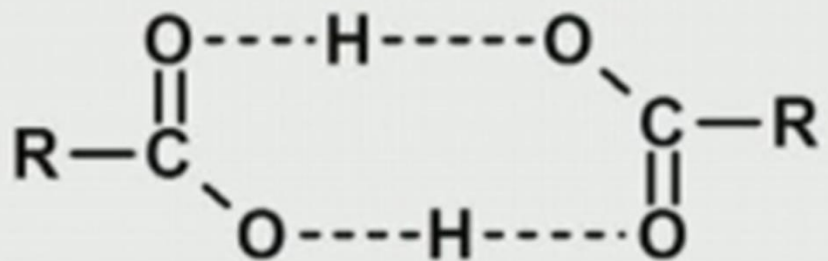


# HYDROGEN BONDS

- The electron density around hydrogen is reduced because of two electronegative groups which are competing for this hydrogen.
- Hence hydrogen atoms involved in hydrogen bonds are typically deshielded.

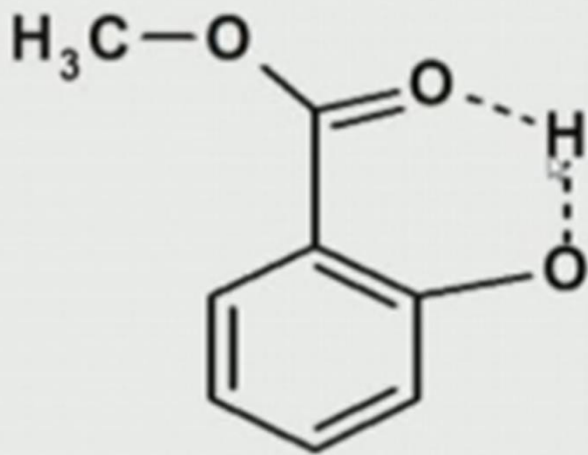


# HYDROGEN BONDS...



Carboxylic acids have strong hydrogen bonding - they form dimers.

With carboxylic acids the O-H absorptions are found between 10 and 12 ppm very far downfield.



In methyl salicylate, which has strong internal hydrogen bonding, the NMR absorption for O-H is at about 14 ppm, way, way downfield.

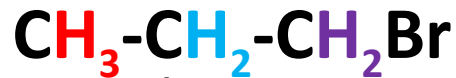
## HYDROGEN BONDS...

- **Molecules having hydrogen bonding have higher chemical shift & absorb radiation at low field.**
- **That is due to the decrease of electronic density around the nucleus**

# $^1\text{H}$ NMR— Number of Signals

## Chemically equivalent protons

- Protons in the same chemical environment
  - Example: the 3 methyl protons in 1-bromo-propane are equiv. because of rotation about C-C bonds

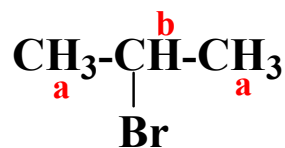


- Each set of chem. equiv. protons in a compound gives rise to a signal in the  $^1\text{H}$ NMR spectrum
  - 1-Bromopropane has **three sets** of chemically chem. equiv. protons, so it has 3 signals in its  $^1\text{H}$  NMR

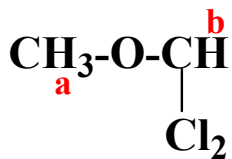
# <sup>1</sup>H NMR—

- From the **no. of signals** in <sup>1</sup>H NMR spectrum, the **kinds of chemically nonequivalent protons** can be known (when there is no overlap)

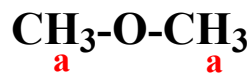
**Example:**



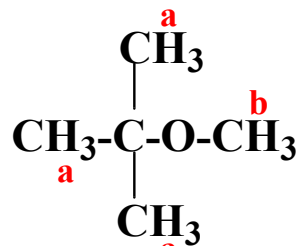
two signals



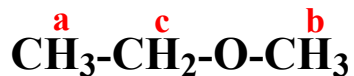
two signals



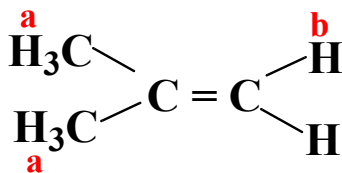
one signal



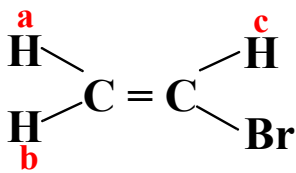
two signals



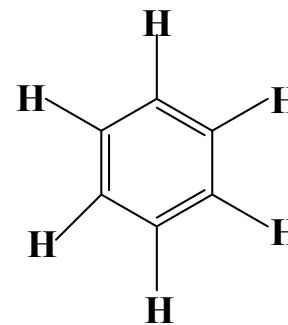
three signals



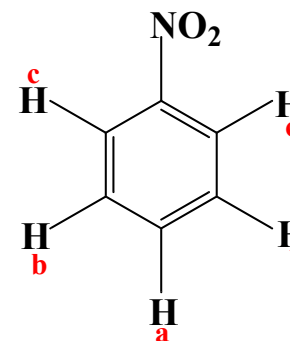
two signals



three signals



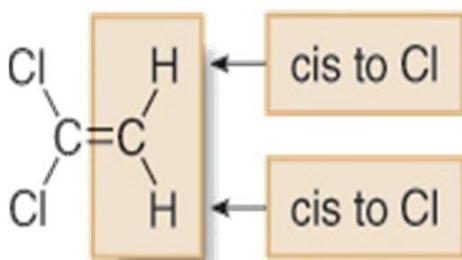
one signal



three signals

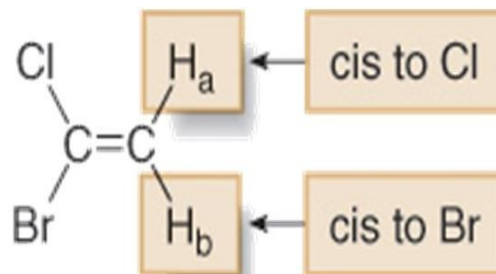
## $^1\text{H}$ NMR—

- In comparing two H atoms on a ring or double bond, two protons are equivalent only if they are cis (or trans) to the same groups.



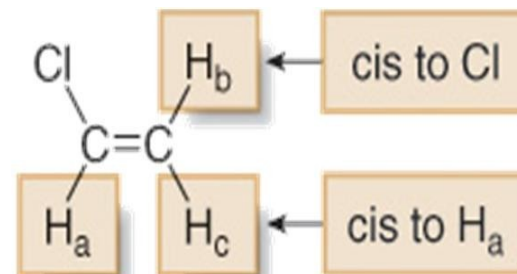
1,1-dichloroethylene

1 type of H  
1 NMR signal



1-bromo-1-chloroethylene

2 types of H's  
2 NMR signals



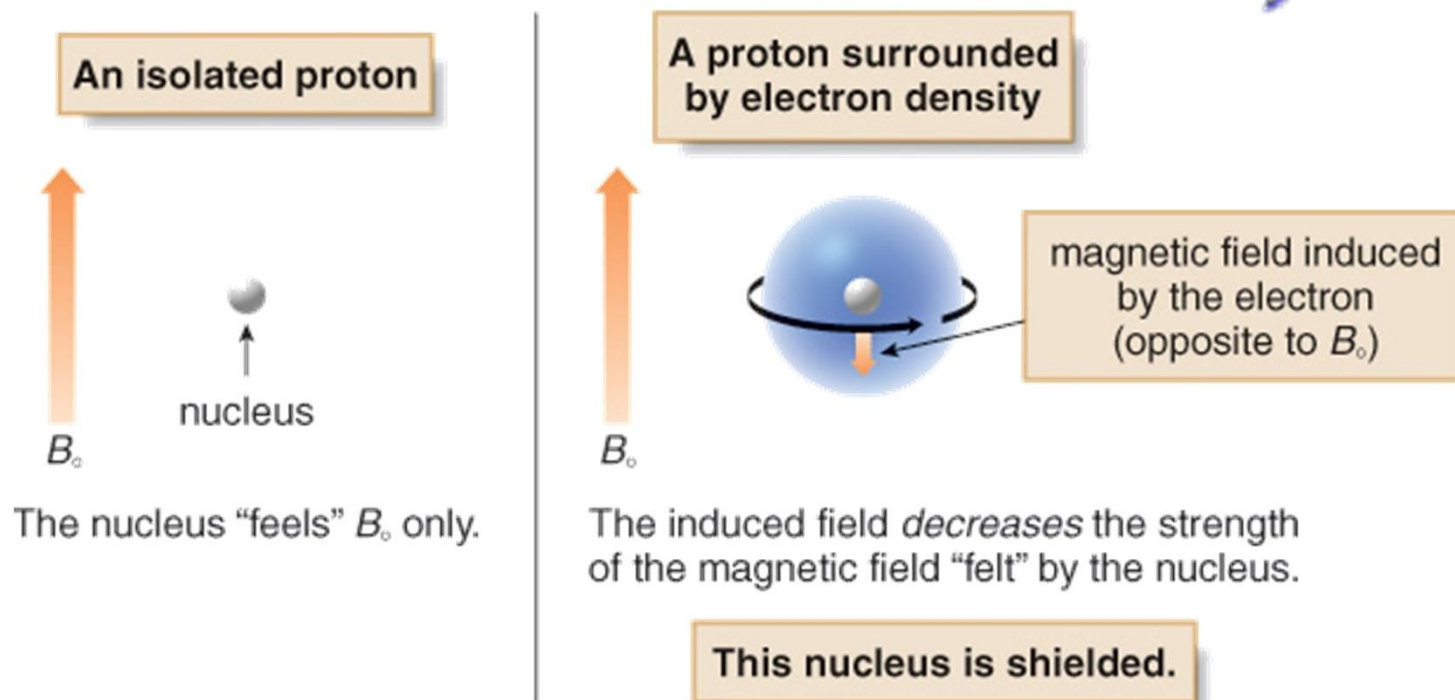
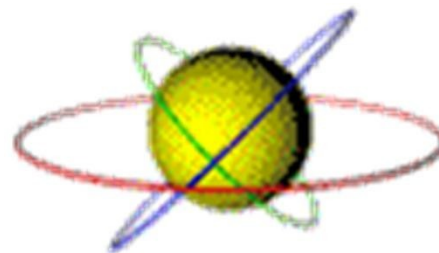
chloroethylene

3 types of H's  
3 NMR signals

## B. $^1\text{H}$ NMR—POSITION OF SIGNALS

- In the vicinity of the nucleus, the magnetic field ( $B_i$ ) generated by the **circulating electron** decreases the external magnetic **field** ( $B_0$ ) that the proton “feels”.

effective field at nucleus:  $B = B_0 - B_i$



# SHEILDING & DESHEILDING EFFECTS

## Highr chemical shift

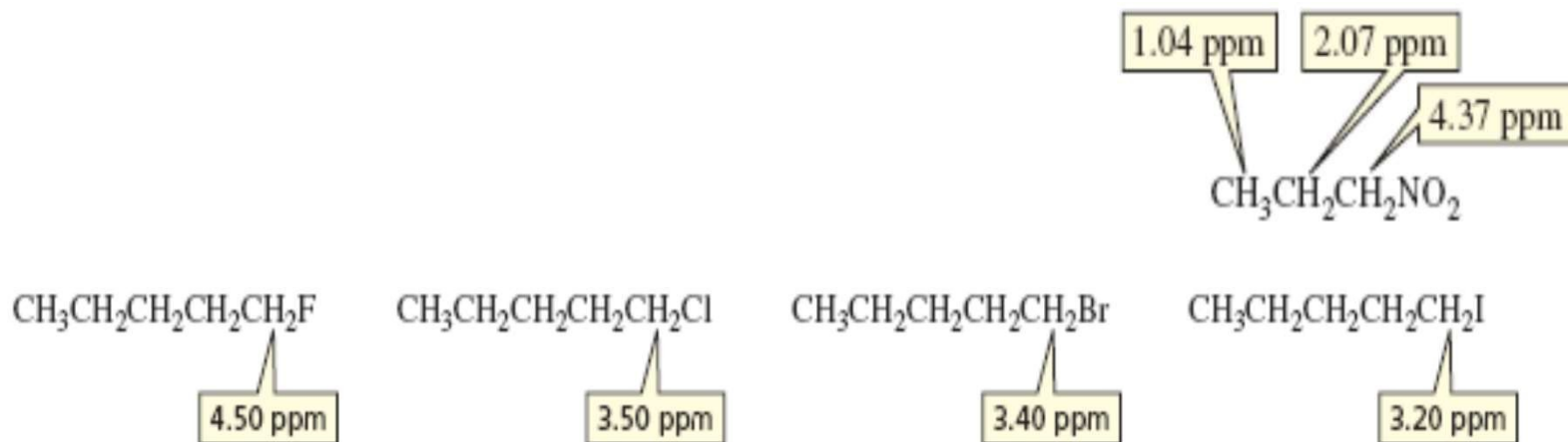
- Downfeild
- The nucleus is more desheilded
- Paramagnetic shift
- Large  $\delta$  value

## Lower chemical shift

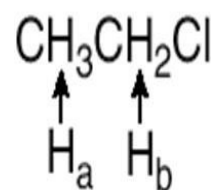
- Upfeild
- The nucleus is more sheilded
- Diamagnetic shift
- Small  $\delta$  value

## THE RELATIVE POSITIONS OF $^1\text{H}$ NMR SIGNALS

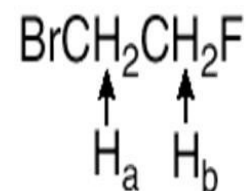
- A nucleus in the vicinity of an electronegative atom experiences reduced electron density & a nucleus is therefore deshielded.
- More electronegative atoms deshield more and give larger shift values.
- Additional electronegative atoms cause increase in chemical shift.



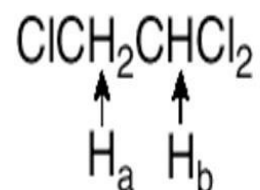
## THE RELATIVE POSITIONS OF $^1\text{H}$ NMR SIGNALS



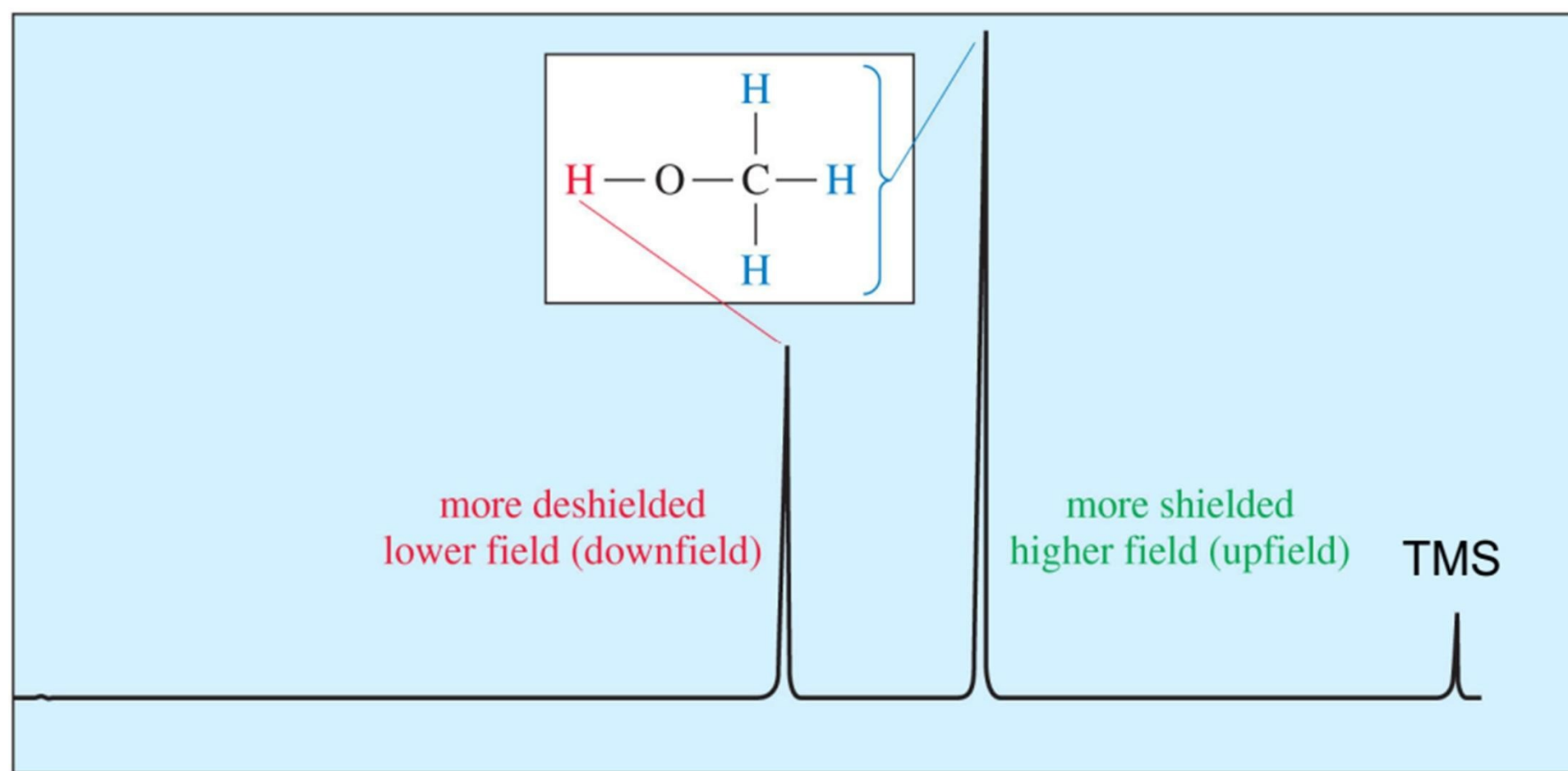
- The  $\text{H}_b$  protons are **deshielded** because they are closer to the electronegative Cl atom, so they absorb **downfield** from  $\text{H}_a$ .



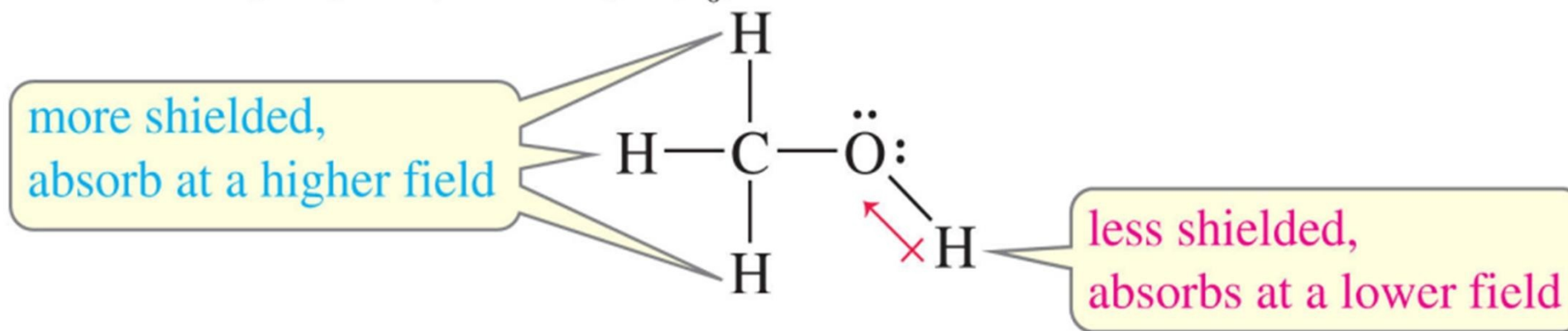
- Because F is more electronegative than Br, the  $\text{H}_b$  protons are more **deshielded** than the  $\text{H}_a$  protons and absorb farther **downfield**.



- The larger number of electronegative Cl atoms (two versus one) **deshields**  $\text{H}_b$  more than  $\text{H}_a$ , so it absorbs **downfield** from  $\text{H}_a$ .

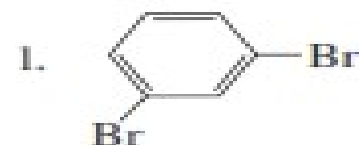
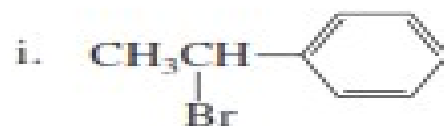


increasing magnetic field strength ( $B_0$ )  $\longrightarrow$

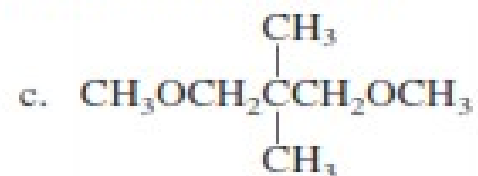


# Exercise

How many signals would you expect to see in the  $^1\text{H}$  NMR spectrum of each of the following compounds?



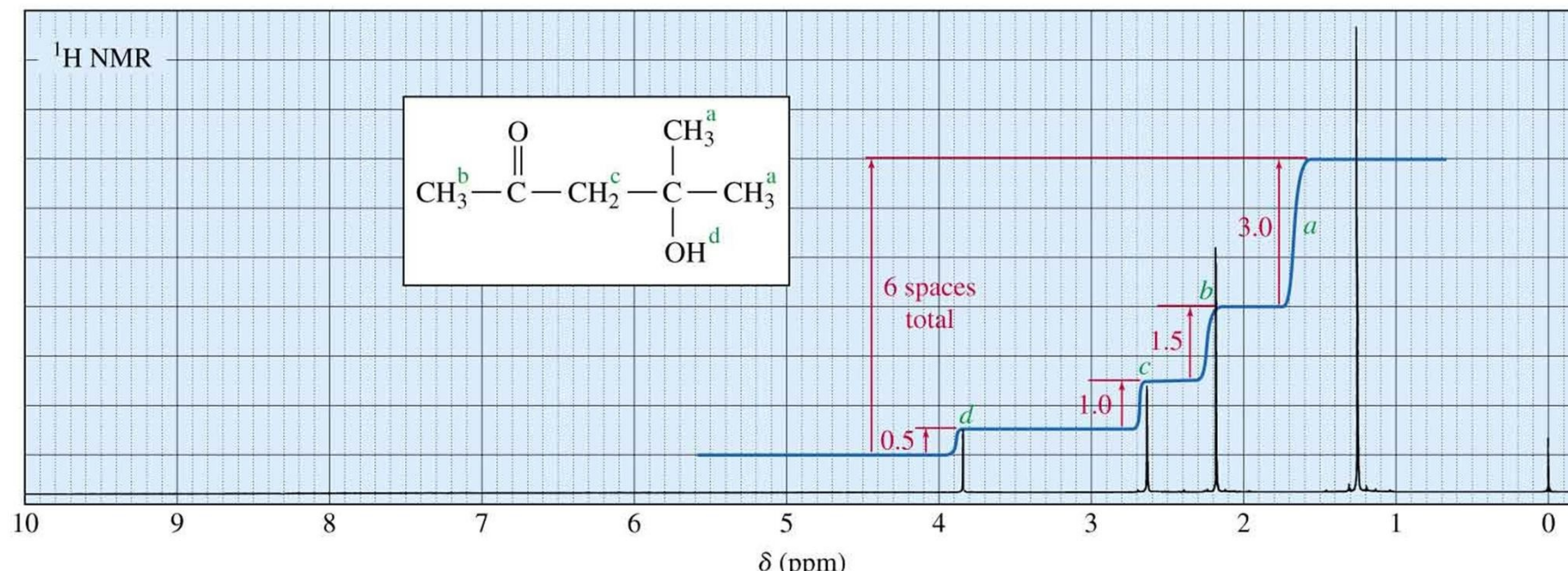
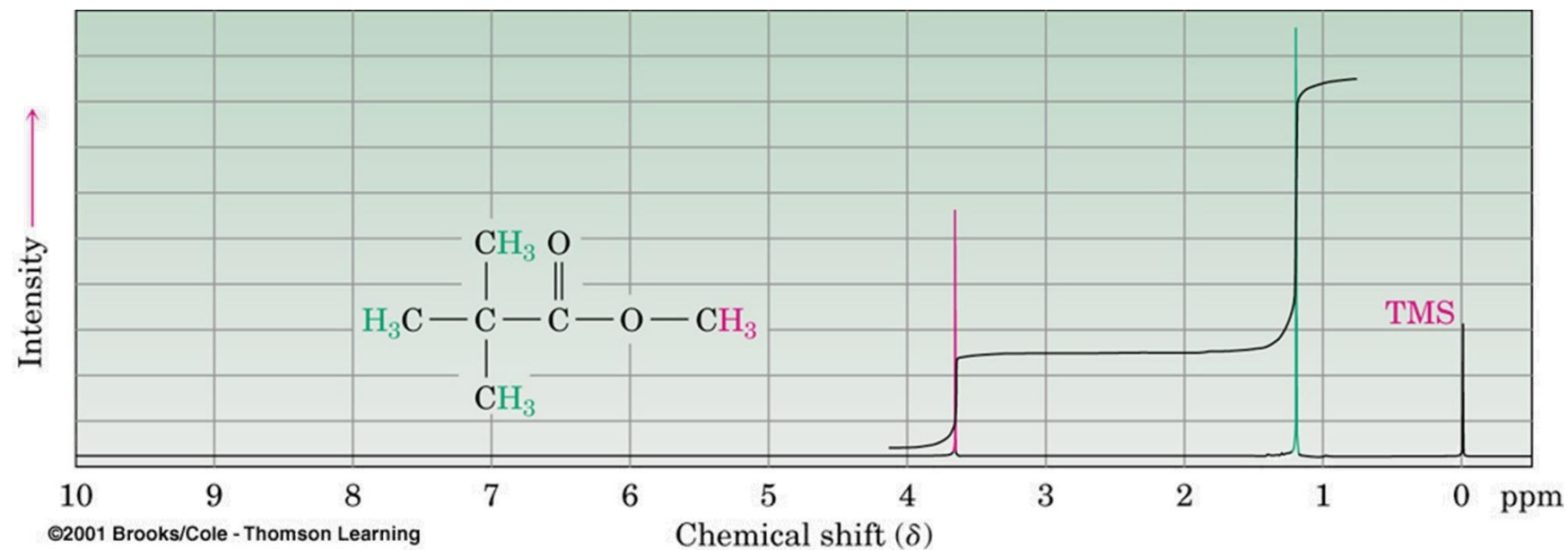
How could you distinguish the  $^1\text{H}$  NMR spectra of the following compounds?



## C. $^1\text{H}$ NMR—Intensity of Signals/integration

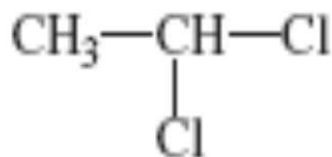
- The area under an NMR signal is proportional to the number of absorbing protons.
- An NMR spectrometer automatically integrates the area under the peaks, and prints out a stepped curve (integral) on the spectrum.
- The height of each step is proportional to the area under the peak, which in turn is proportional to the number of absorbing protons.
- Modern NMR spectrometers automatically calculate and plot the value of each integral in arbitrary units.
- The ratio of integrals to one another gives the ratio of absorbing protons in a

## C. $^1\text{H}$ NMR—Intensity of Signals/integration

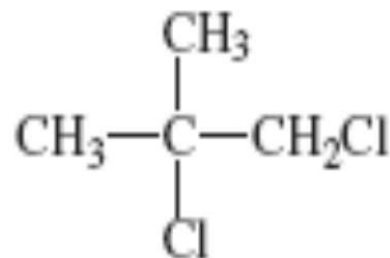


## $^1\text{H}$ NMR—Intensity of Signals...

- The integration tells us the *relative* number of protons that give rise to each signal, not the *absolute* number.
- For example, integration could not distinguish between 1,1- dichloroethane and 1,2-dichloro-2-methylpropane because both compounds would show an integral ratio of 1:3.



1,1-dichloroethane  
ratio of protons = 1:3

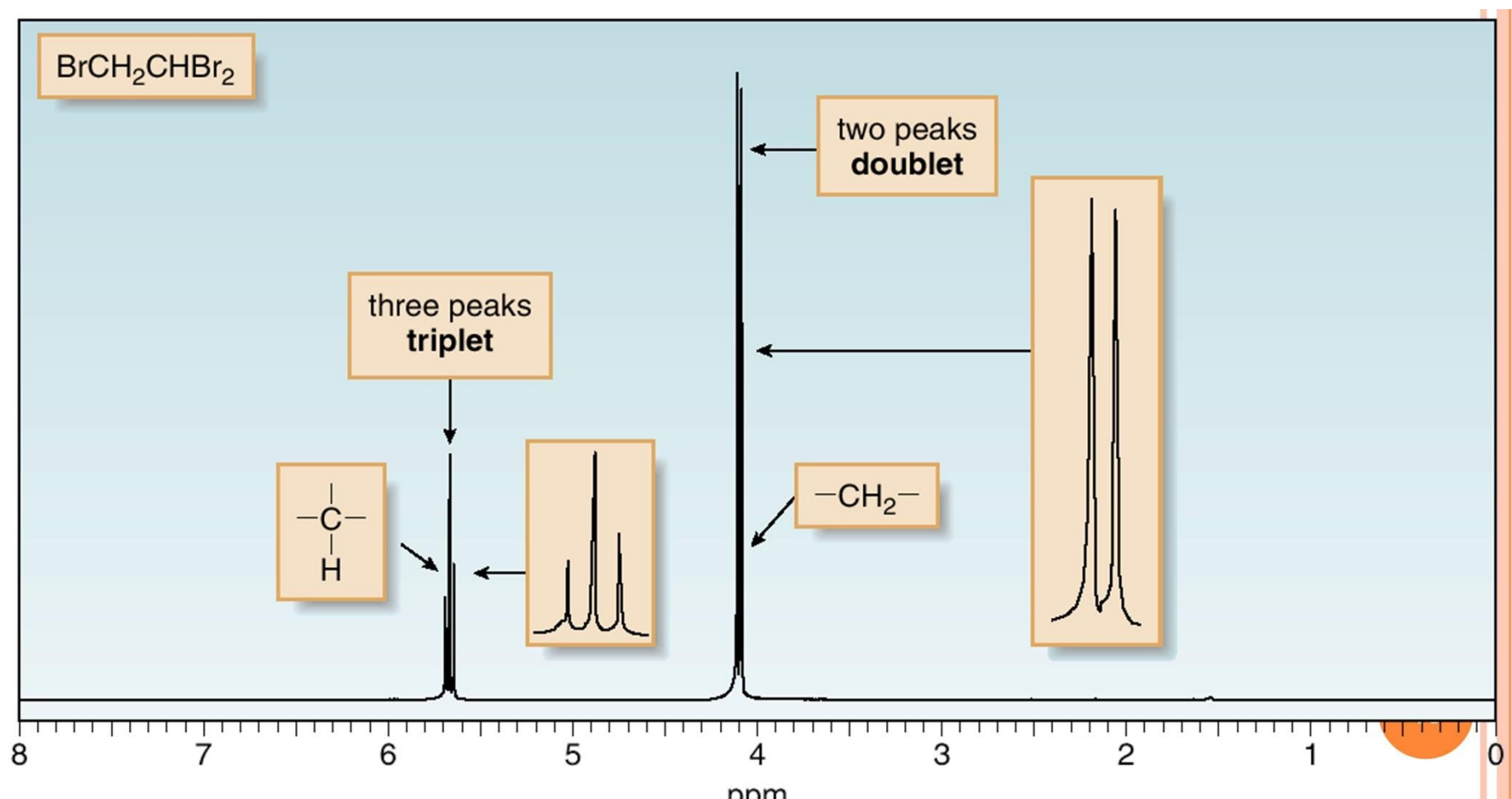


1,2-dichloro-2-methylpropane  
ratio of protons 2:6 = 1:3

# $^1\text{H}$ NMR—Spin-Spin Splitting

- Consider the spectrum below:

Happens because neighboring hydrogen atoms (protons) influence each other's magnetic environment.



## SPIN-SPIN SPLITTING IN $^1\text{H}$ NMR SPECTRA

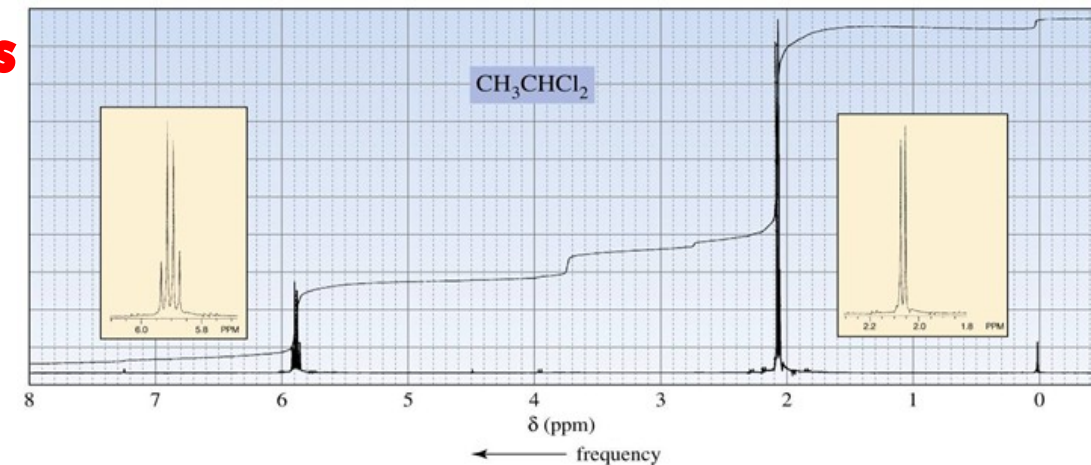
- Peaks are often split into multiple peaks due to *magnetic interactions* between nonequivalent protons on adjacent carbons, the process is called **spin-spin splitting**.
- The splitting is into one more peak than the number of H's on the adjacent carbon(s).
- This is the **"n+1 rule"**
- The relative intensities are in proportion of a binomial distribution given by Pascal's Triangle
- The set of peaks is a **multiplet** (2 = doublet, 3 = triplet, 4 = quartet, 5=pentet, 6=hextet, 7=heptet.....)

- fore example 1,1-dichloroethane
  - The peak farthest up field is split into 2 peaks (a doublet) and the methine proton is split into 4 peaks (a quartet)

- The above 2 sets of protons are called **coupled protons**
- according to the **"n+1 rule"**  $\text{CH}_3^{\text{a}}\text{-CH}^{\text{b}}\text{Cl}_2$

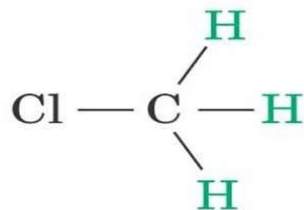
a is split into (N+1); 1+1 = 2 doublet

b is split into (N+3); 1+3 = 4 quartet



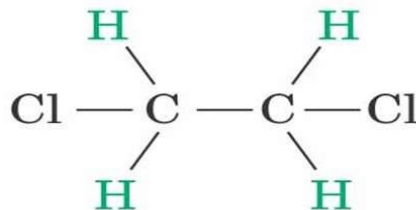
## Rules for Spin-Spin Splitting

- **Equivalent** protons **do not** split each other



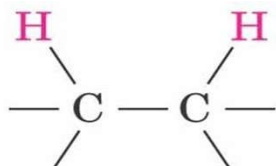
Three C–H protons are chemically equivalent; no splitting occurs.

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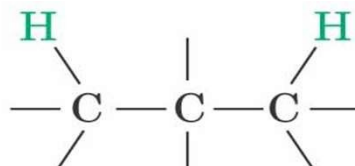
Four C–H protons are chemically equivalent; no splitting occurs.

- **Protons that are farther than two carbon atoms apart do not split each other**



Splitting observed

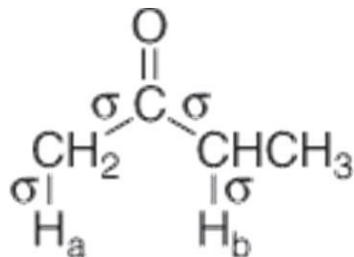
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Splitting not usually observed

# Rules for Spin-Spin Splitting....

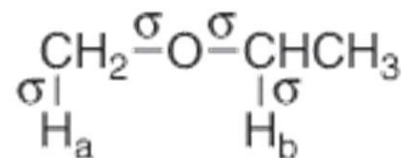
- **Splitting is not generally observed between protons separated by more than three  $\sigma$  bonds.**



2-butanone

$H_a$  and  $H_b$  are separated by four  $\sigma$  bonds.

**no splitting** between  $H_a$  and  $H_b$



ethyl methyl ether

$H_a$  and  $H_b$  are separated by four  $\sigma$  bonds.

**no splitting** between  $H_a$  and  $H_b$

# The ( $n + 1$ ) Rule

- If a hydrogen atom has  $n$  neighboring hydrogens, its signal will split into  $n + 1$  peaks.
- Examples:
  - I. 0 neighboring H  $\rightarrow$  1 peak (singlet)
  - II. 1 neighboring H  $\rightarrow$  2 peaks (doublet)
  - III. 2 neighboring H  $\rightarrow$  3 peaks (triplet)
  - IV. 3 neighboring H  $\rightarrow$  4 peaks (quartet)

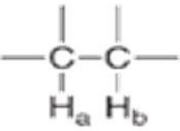
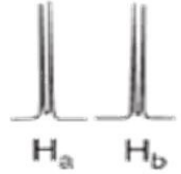
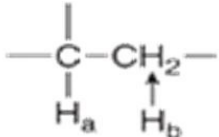
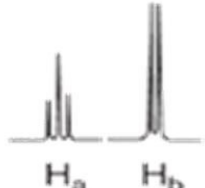
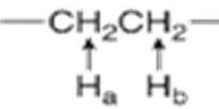
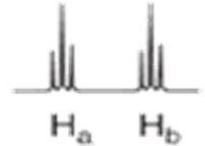
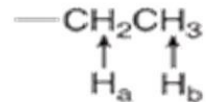
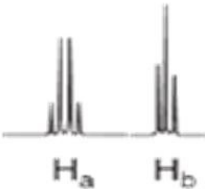
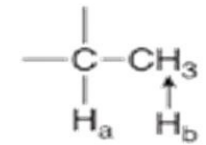
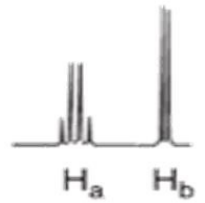
## Splitting pattern: how many neighboring hydrogens.

In general, n-equivalent neighboring hydrogens will split a  $^1\text{H}$  signal into an  $(n + 1)$  Pascal pattern.

“neighboring” – no more than three bonds away

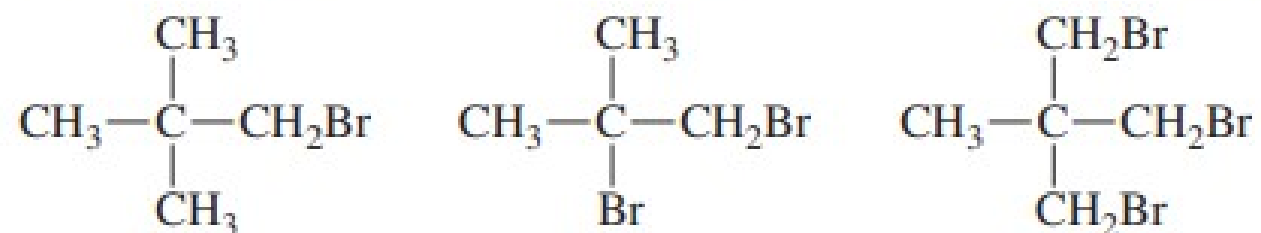
| n | n + 1 | Pascal pattern: |         |
|---|-------|-----------------|---------|
| 0 | 1     | 1               | singlet |
| 1 | 2     | 1 1             | doublet |
| 2 | 3     | 1 2 1           | triplet |
| 3 | 4     | 1 3 3 1         | quartet |

**Table 14.4****Common Splitting Patterns Observed in  $^1\text{H}$  NMR**

| Example                                                                                 | Pattern                                                                             | Analysis ( $\text{H}_a$ and $\text{H}_b$ are not equivalent.)                                                                                                                                                                                              |             |               |                   |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|-------------------|
| [1]    |    | <ul style="list-style-type: none"> <li><math>\text{H}_a</math>: one adjacent <math>\text{H}_b</math> proton <math>\rightarrow</math></li> <li><math>\text{H}_b</math>: one adjacent <math>\text{H}_a</math> proton <math>\rightarrow</math></li> </ul>     | two peaks   | $\rightarrow$ | a <b>doublet</b>  |
| [2]    |    | <ul style="list-style-type: none"> <li><math>\text{H}_a</math>: two adjacent <math>\text{H}_b</math> protons <math>\rightarrow</math></li> <li><math>\text{H}_b</math>: one adjacent <math>\text{H}_a</math> proton <math>\rightarrow</math></li> </ul>    | three peaks | $\rightarrow$ | a <b>triplet</b>  |
| [3]    |    | <ul style="list-style-type: none"> <li><math>\text{H}_a</math>: two adjacent <math>\text{H}_b</math> protons <math>\rightarrow</math></li> <li><math>\text{H}_b</math>: two adjacent <math>\text{H}_a</math> protons <math>\rightarrow</math></li> </ul>   | three peaks | $\rightarrow$ | a <b>triplet</b>  |
| [4]   |   | <ul style="list-style-type: none"> <li><math>\text{H}_a</math>: three adjacent <math>\text{H}_b</math> protons <math>\rightarrow</math></li> <li><math>\text{H}_b</math>: two adjacent <math>\text{H}_a</math> protons <math>\rightarrow</math></li> </ul> | four peaks  | $\rightarrow$ | a <b>quartet*</b> |
| [5]  |  | <ul style="list-style-type: none"> <li><math>\text{H}_a</math>: three adjacent <math>\text{H}_b</math> protons <math>\rightarrow</math></li> <li><math>\text{H}_b</math>: one adjacent <math>\text{H}_a</math> proton <math>\rightarrow</math></li> </ul>  | four peaks  | $\rightarrow$ | a <b>quartet*</b> |
|                                                                                         |                                                                                     |                                                                                                                                                                                                                                                            | two peaks   | $\rightarrow$ | a <b>doublet</b>  |

\*The relative area under the peaks of a quartet is 1:3:3:1.

How would integration distinguish the  $^1\text{H}$  NMR spectra of the following compounds?



Indicate the number of signals and the multiplicity of each signal in the  $^1\text{H}$  NMR spectrum of each of the following compounds:



$^{13}\text{C}$  - NMR

## What can we deduce about molecular structure from $^{13}\text{C}$ -NMR spectrum?

- NMR fundamentals are the same regardless of nucleus

### Information from carbon NMR spectrum

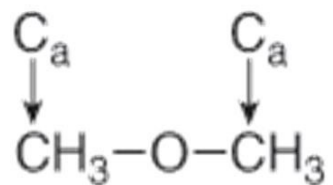
- Number of signals: equivalent carbons and molecular symmetry
- Chemical shift: presence of high EN atoms or pi electron clouds

The two features of a  $^{13}\text{C}$  NMR spectrum that provide the most structural information are the **number of signals** observed and the **chemical shifts** of those signals.

## $^{13}\text{C}$ NMR—Number of Signals

- The number of signals in a  $^{13}\text{C}$  spectrum gives the number of different types of carbon atoms in a molecule.
- Because  $^{13}\text{C}$  NMR signals are not split, the number of signals equals the number of lines in the  $^{13}\text{C}$  spectrum.
- In contrast to the  $^1\text{H}$  NMR situation, peak intensity is not proportional to the number of absorbing carbons, so  $^{13}\text{C}$  NMR signals are not integrated.

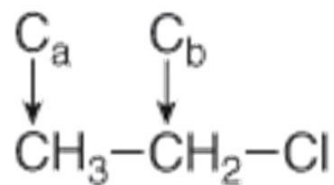
## $^{13}\text{C}$ NMR...



dimethyl ether

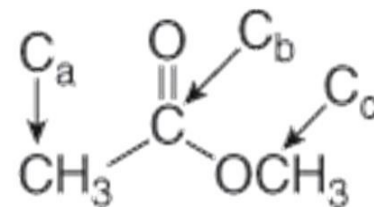
1  $^{13}\text{C}$  NMR signal

Both C's are equivalent.



chloroethane

2  $^{13}\text{C}$  NMR signals



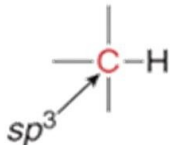
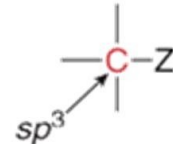
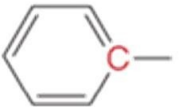
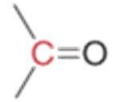
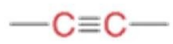
methyl acetate

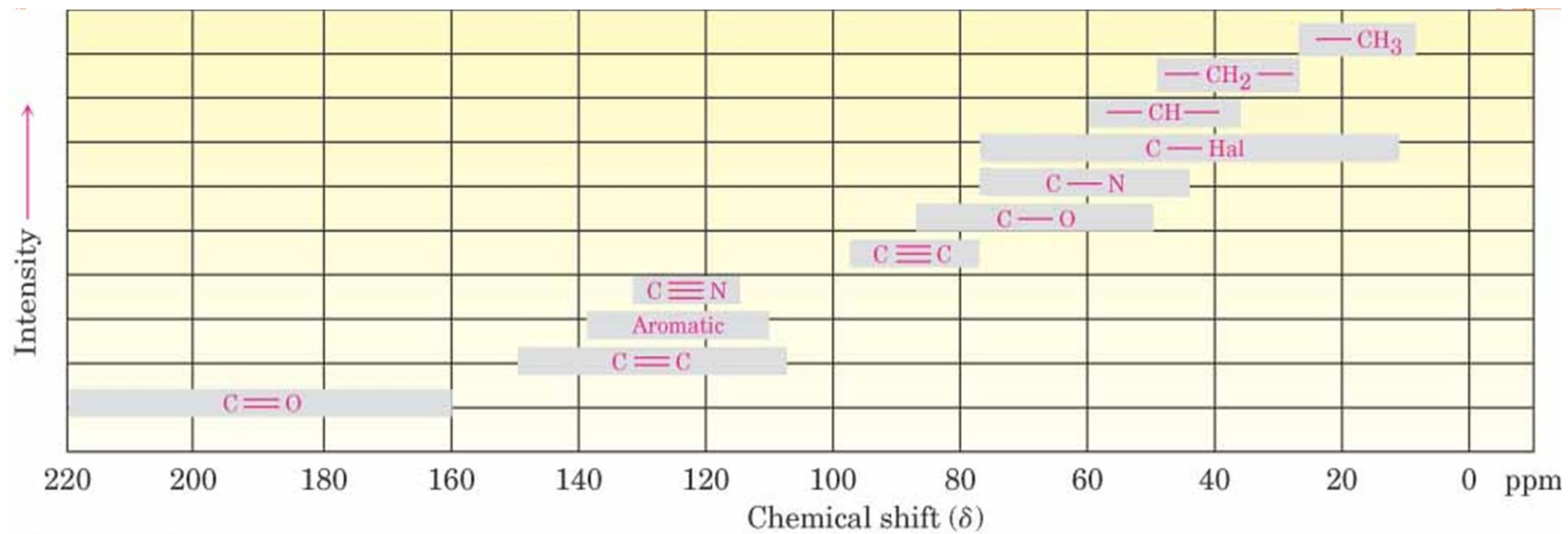
3  $^{13}\text{C}$  NMR signals

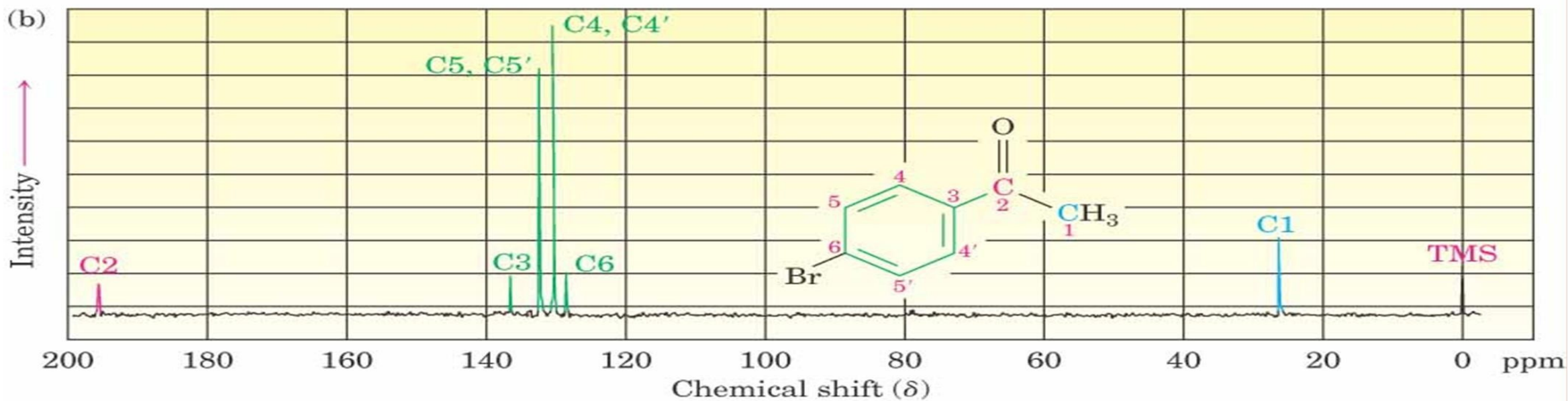
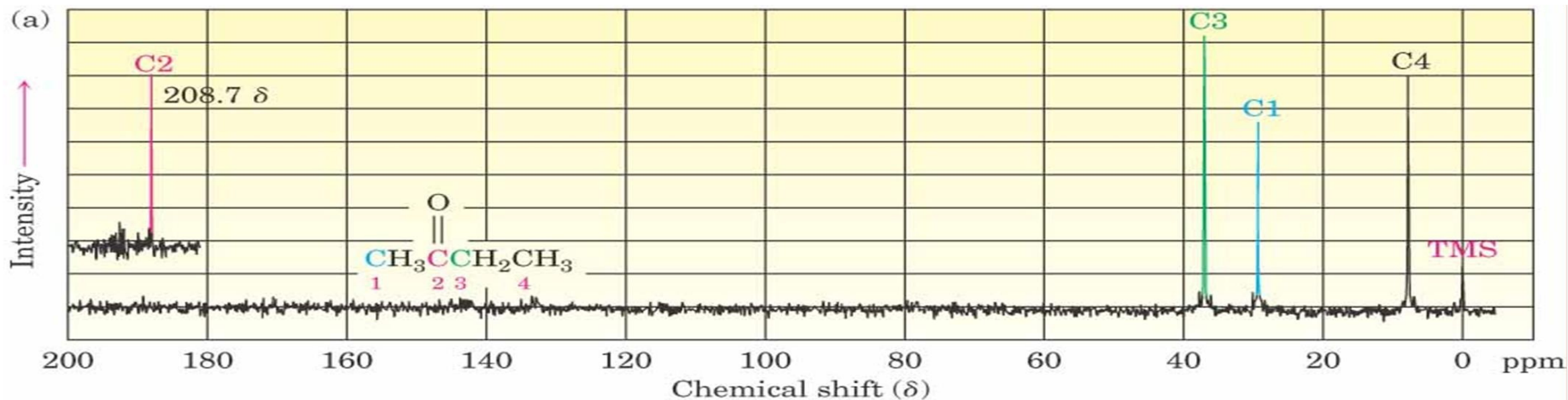
## $^{13}\text{C}$ NMR...

### $^{13}\text{C}$ NMR—Position of Signals

- In contrast to the small range of chemical shifts in  $^1\text{H}$  NMR (1-10 ppm usually),  $^{13}\text{C}$  NMR absorptions occur over a much broader range (0-220 ppm).
- The chemical shifts of carbon atoms in  $^{13}\text{C}$  NMR depend on the same effects as the chemical shifts of protons in  $^1\text{H}$  NMR.

| Type of carbon                                                                      | Chemical shift (ppm) | Type of carbon                                                                        | Chemical shift (ppm) |
|-------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------|----------------------|
|   | 5–45                 |   | 100–140              |
|  | 30–80                |  | 120–150              |
| Z = N, O, X                                                                         |                      |  | 160–210              |
|  | 65–100               |                                                                                       |                      |

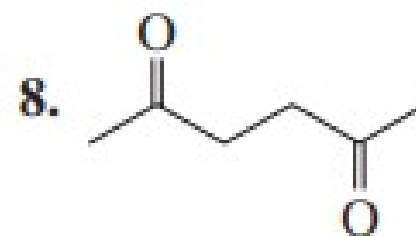
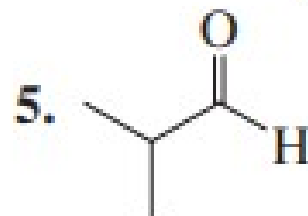
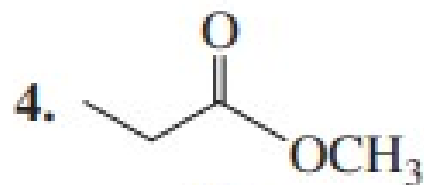
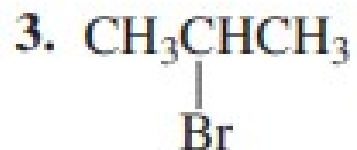




Answer the following questions for each of the following compounds:

a. How many signals are in its  $^{13}\text{C}$  NMR spectrum?

b. Which signal is at the lowest frequency?



Thank you